

PERIODIC FACILITY REVIEW

GEOTECHNICAL INVESTIGATION REPORT

Elk Basin Dam

FERC Project No. 4182

Elk Basin Irrigation Project

Elk Creek, Elmore County, Idaho

Prepared for:

Elk Basin Irrigation District

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1.0 INTRODUCTION

1.1 Purpose and Scope

Mountain West Dam Engineering (MWDE) was retained by the Elk Basin Irrigation District (EBID) to perform a geotechnical investigation as part of the Periodic Facility Review (PFR) for Elk Basin Dam. The investigation addresses three conditions identified during the 2020 FERC Part 12D independent consultant's safety inspection (Cascade Dam Safety Consultants, October 2020): (1) alkali-silica reaction (ASR) in the arch dam concrete, (2) increasing seepage at the left abutment, and (3) horizontal displacement of the right abutment embankment crest.

The scope of services included drilling and sampling of three exploratory borings, comprehensive laboratory testing, engineering analysis, and development of conclusions and recommendations. This report presents the investigation findings and provides the data required for EBID and its dam safety consultants to evaluate remedial alternatives.

1.2 Background

Elk Basin Dam is a 145-foot-high concrete arch dam with earthen embankment abutment transitions, constructed by the U.S. Bureau of Reclamation between 1966 and 1968. The dam impounds 18,500 acre-feet on Elk Creek for agricultural irrigation and hydroelectric power generation. The dam is founded on Cretaceous-age biotite granodiorite of the Idaho Batholith. The embankment abutments are founded on a combination of granodiorite bedrock and dense Quaternary glacial till. Comprehensive construction and design data are documented in the USBR As-Built Construction Report (December 1968).

The 2020 FERC inspection rated the overall dam condition as FAIR and recommended this PFR investigation to characterize the three identified conditions. Specifically, the inspection recommended: (a) concrete core drilling through the arch to assess ASR depth and residual strength; (b) a boring through the left abutment embankment to investigate the seepage path; and (c) a boring through the right abutment embankment to characterize foundation conditions for stability analysis.

2.0 FIELD EXPLORATION

2.1 Drilling Program

Three exploratory borings were drilled between August 18 and August 29, 2025. Drilling was performed by Idaho Core Drilling, Inc. of Boise, Idaho. All borings were logged by MWDE geotechnical engineers. The boring program is summarized in the following table.

Table 2-1: Boring Program Summary

Boring	Location	Ground Elev. (ft)	Total Depth (ft)	GW (ft bgs)	Method	Instruments Installed
BH-LA-01	Left abt. crest, Sta. 2+40	6,238.5	120.0	42.0	HSA / HQ core	VW piezometer at 95 ft
BH-RA-01	Right abt. mid-slope, Sta. 6+80	6,210.3	95.0	35.0	HSA / HQ core	VW piezometer at 75 ft; Inclinator to 90 ft
BH-AC-01	Arch crest, Sta. 4+60 (crown)	6,245.0	28.0	Seepage at joints	NX diamond core (continuous)	None

2.2 Left Abutment Boring (BH-LA-01)

BH-LA-01 was drilled from the left abutment embankment crest at Station 2+40. The boring was advanced through the embankment fill zones using 4.25-inch hollow-stem augers with split-spoon and Shelby tube sampling. Upon reaching the original ground surface at approximately 65 feet, drilling continued through alluvial deposits and into bedrock using HQ wireline diamond coring. The boring was terminated at 120 feet in fresh granodiorite.

The subsurface profile encountered in BH-LA-01 is: (1) embankment gravel shell, 0 to 25 feet; (2) embankment clay core, 25 to 45 feet; (3) embankment gravel shell / filter, 45 to 65 feet; (4) original ground surface / alluvial deposits, 65 to 72 feet; (5) weathered granodiorite (grus), 72 to 85 feet; (6) fresh granodiorite bedrock, 85 to 120 feet. The embankment clay core appeared intact with no evidence of internal erosion or piping. A thin (0.5-inch) clay gouge layer was observed at the foundation contact (65 feet).

Groundwater was measured at 42.0 feet below the embankment crest, within the embankment above the clay core. This is consistent with the design phreatic surface. The vibrating wire piezometer installed at 95 feet (in fresh granodiorite) recorded an initial reading of 18.2 psi, indicating a piezometric head approximately 42 feet above the sensor. This elevated pore pressure in the foundation rock confirms a seepage path through the weathered granodiorite zone.

2.3 Right Abutment Boring (BH-RA-01)

BH-RA-01 was drilled from the right abutment embankment mid-slope at Station 6+80, near the maximum embankment section where the greatest displacement has been measured. The boring was advanced through embankment fill using hollow-stem augers with sampling, then continued into the glacial till foundation and underlying bedrock using HQ wireline coring.

The subsurface profile: (1) embankment gravel shell, 0 to 15 feet; (2) embankment clay core, 15 to 32 feet; (3) embankment gravel shell / filter, 32 to 48 feet; (4) glacial till, 48 to 55 feet; (5) weathered granodiorite, 55 to 70 feet; (6) fresh granodiorite, 70 to 95 feet. The glacial till is a very dense, heterogeneous clayey gravel with cobbles, classified as GC. The till/bedrock contact at 55 feet is gradational over approximately 2 feet.

An inclinometer casing was installed to 90 feet to enable future monitoring of the displacement profile with depth. Initial readings establish the baseline for comparison with future measurements. The vibrating wire piezometer at 75 feet recorded 12.8 psi, indicating moderate but stable pore pressure in the foundation rock.

2.4 Arch Concrete Boring (BH-AC-01)

BH-AC-01 was drilled vertically from the arch dam crest at Station 4+60 (near the arch crown) using NX-size (3-inch core diameter) continuous diamond coring. The boring penetrated 22.0 feet of mass concrete (the full arch thickness at this location) and continued 6.0 feet into the granodiorite foundation rock. Core recovery in the concrete ranged from 92 to 98 percent, with reduced recovery in the upper 4 feet where ASR cracking was most severe.

Visible ASR indicators (gel deposits, reaction rims on aggregate particles, and microcracking) were observed in the core from the surface to approximately 6 feet depth. Below 6 feet, the concrete appeared visually sound with no ASR indicators. The concrete/rock foundation contact at 22.0 feet was intact with good bond and no deterioration. The contact grout installed during construction was observed and appears sound. Minor seepage was noted at two construction lift joints (at approximately 7 and 12 feet depth) during coring.

3.0 LABORATORY TESTING

Laboratory testing was performed by MWDE's geotechnical laboratory in Meridian, Idaho (rock mechanics and concrete testing), and by Intermountain Analytical Laboratories of Grand Junction, Colorado (soil classification and index testing). All testing was performed in accordance with the applicable ASTM standards.

3.1 Embankment Fill Testing

3.1.1 Clay Core — Index Properties

Table 3-1: Embankment Clay Core — Index Properties and Density

Sample ID	Boring	Depth (ft)	Zone	LL	PL	PI	USCS	MC (%)	Dry Den. (pcf)	Rel. Comp. (%)
LA-01-SS8	BH-LA-01	28-30	Core	42	22	20	CL	24.3	107.5	94.5
LA-01-SS10	BH-LA-01	35-37	Core	44	23	21	CL	25.1	106.8	93.9
LA-01-SS12	BH-LA-01	40-42	Core	41	21	20	CL	23.8	108.0	94.9
RA-01-SS5	BH-RA-01	18-20	Core	40	21	19	CL	23.1	108.8	95.0
RA-01-SS7	BH-RA-01	25-27	Core	39	20	19	CL	22.8	109.2	95.4
RA-01-SS9	BH-RA-01	30-32	Core	41	22	19	CL	24.0	107.8	94.2

The clay core material classifies as CL (lean clay) in both abutments, consistent with the original construction records. Liquid limits range from 39 to 44, plasticity indices from 19 to 21. Relative compaction values of 93.9 to 95.4 percent indicate the core was placed at or above the specified minimum of 93 percent. Moisture contents are 1 to 3 percent above the plastic limit, indicating the core is near saturation as expected for a clay core under long-term reservoir loading.

3.1.2 Clay Core — Gradation

Table 3-2: Embankment Clay Core — Gradation with Hydrometer

Sample ID	Boring	Depth (ft)	Gravel (%)	Sand (%)	Silt (%)	Clay (%)	D60 (mm)	Activity
LA-01-SS8	BH-LA-01	28-30	3.2	26.8	42.5	27.5	0.038	0.73
LA-01-SS12	BH-LA-01	40-42	2.8	28.2	41.0	28.0	0.042	0.71
RA-01-SS5	BH-RA-01	18-20	3.5	29.5	40.8	26.2	0.048	0.73
RA-01-SS9	BH-RA-01	30-32	2.1	27.9	43.2	26.8	0.035	0.71

3.1.3 Clay Core — Triaxial Strength

Table 3-3: Embankment Clay Core — Consolidated Undrained Triaxial (ASTM D4767)

Sample ID	Boring	Depth (ft)	Test Type	Cell Press. (psi)	c' (psf)	phi' (deg)	Notes
LA-01-ST1	BH-LA-01	33-35	CU w/ pp	15, 30, 60	320	24	Intact, no softening
RA-01-ST1	BH-RA-01	22-24	CU w/ pp	15, 30, 60	290	23	Intact, slight softening at high strain

The effective stress strength parameters ($c' = 290\text{--}320$ psf, $\phi' = 23\text{--}24$ degrees) are consistent with published values for compacted lean clay and indicate the core material has maintained its design strength over 57 years of service. No evidence of progressive softening, dispersivity, or internal erosion was observed in any core

sample.

3.2 Foundation Soil Testing

3.2.1 Glacial Till (Right Abutment Foundation)

Table 3-4: Glacial Till Foundation — Index Properties

Sample ID	Boring	Depth (ft)	LL	PL	PI	USCS	MC (%)	Gravel (%)	Sand (%)	Fines (%)
RA-01-SS14	BH-RA-01	49-51	28	16	12	GC	11.2	52	28	20
RA-01-SS15	BH-RA-01	52-54	26	15	11	GC	10.8	48	31	21

Table 3-5: Glacial Till — Consolidated Undrained Triaxial

Sample ID	Boring	Depth (ft)	Test Type	Cell Press. (psi)	c' (psf)	phi' (deg)	Notes
RA-01-UD1	BH-RA-01	50-52	CU w/ pp	30, 60, 120	450	34	Very dense, well-graded, slow creep at high deviator stress

The glacial till exhibited measurable creep strain during the sustained-loading phase of the highest confining pressure test (120 psi). Axial strain increased at approximately 0.002 percent per hour under a deviator stress of 85 percent of the peak value. This creep behavior is consistent with the observed constant-rate downstream displacement of the right abutment embankment and supports the hypothesis that the displacement is attributable to foundation creep rather than progressive failure.

3.3 Rock Strength Testing

3.3.1 Unconfined Compressive Strength — Fresh Granodiorite

Table 3-6: UCS Results — Fresh Granodiorite (ASTM D7012 Method C)

Specimen ID	Boring	Depth (ft)	D (in)	L (in)	UCS (psi)	E50 (ksi)	nu	Failure Mode
LA-01-RC2-U1	BH-LA-01	88-89	2.13	4.45	21,200	7,850	0.19	Cone/shear
LA-01-RC2-U2	BH-LA-01	95-96	2.13	4.48	24,800	8,620	0.17	Axial split
LA-01-RC3-U1	BH-LA-01	105-106	2.13	4.42	22,100	8,100	0.18	Shear 55 deg
LA-01-RC3-U2	BH-LA-01	112-113	2.13	4.50	25,600	9,180	0.17	Cone/shear
RA-01-RC2-U1	BH-RA-01	72-73	2.13	4.44	18,400	6,520	0.21	Axial split
RA-01-RC2-U2	BH-RA-01	78-79	2.13	4.47	21,800	7,680	0.19	Shear 60 deg
RA-01-RC3-U1	BH-RA-01	85-86	2.13	4.43	23,500	8,400	0.18	Cone/shear
RA-01-RC3-U2	BH-RA-01	90-91	2.13	4.46	20,600	7,200	0.20	Axial (joint)
AC-01-RC1-U1	BH-AC-01	24-25	2.13	4.48	24,200	8,800	0.17	Cone/shear

UCS values range from 18,400 to 25,600 psi with a mean of 22,467 psi and standard deviation of 2,290 psi. These values are consistent with the 1964 design investigation data (mean 22,100 psi) and confirm the foundation rock strength has not degraded. E50 ranges from 6,520 to 9,180 ksi (mean 8,039 ksi). Poisson's ratio ranges from 0.17 to 0.21 (mean 0.18). The lowest UCS value (18,400 psi, BH-RA-01 at 72 ft) corresponds to the shallowest bedrock specimen, which showed slight weathering effects.

3.3.2 Unconfined Compressive Strength — Weathered Granodiorite

Table 3-7: UCS Results — Weathered Granodiorite

Specimen ID	Boring	Depth (ft)	UCS (psi)	Weathering Grade	Notes
LA-01-RC1-W1	BH-LA-01	74-75	2,650	Highly (HW)	Friable, grus texture
LA-01-RC1-W2	BH-LA-01	78-79	1,820	Highly (HW)	Sandy decomposition
LA-01-RC1-W3	BH-LA-01	82-83	4,100	Moderately (MW)	Transitional to fresh
RA-01-RC1-W1	BH-RA-01	58-59	1,280	Completely (CW)	Soil-like, hand crumbles
RA-01-RC1-W2	BH-RA-01	63-64	2,450	Highly (HW)	Relict texture preserved
RA-01-RC1-W3	BH-RA-01	67-68	3,800	Moderately (MW)	Transitional

The weathered granodiorite (grus) shows a clear strength gradient from completely weathered (1,280 psi at the till contact) to moderately weathered (3,800-4,100 psi) at the transition to fresh rock. The elevated permeability of this zone (estimated 10 to 100 times that of fresh granodiorite based on Lugeon test correlations) is the likely pathway for the increased seepage observed at the left abutment.

3.3.3 Indirect Tensile Strength — Fresh Granodiorite

Table 3-8: ITS Results — Fresh Granodiorite (ASTM D3967)

Specimen ID	Boring	Depth (ft)	D (in)	t (in)	Peak (lbf)	ITS (psi)	Failure
LA-01-RC2-T1	BH-LA-01	89-90	2.13	1.05	3,480	1,580	Central
LA-01-RC2-T2	BH-LA-01	92-93	2.13	1.02	3,120	1,460	Central
LA-01-RC2-T3	BH-LA-01	96-97	2.13	1.08	2,850	1,260	Off-center
LA-01-RC3-T1	BH-LA-01	108-109	2.13	1.04	3,650	1,680	Central

Specimen ID	Boring	Depth (ft)	D (in)	t (in)	Peak (lbf)	ITS (psi)	Failure
LA-01-RC3-T2	BH-LA-01	114-115	2.13	1.06	3,380	1,520	Central
RA-01-RC2-T1	BH-RA-01	74-75	2.13	1.03	2,680	1,240	Central
RA-01-RC2-T2	BH-RA-01	80-81	2.13	1.05	3,210	1,460	Central
RA-01-RC3-T1	BH-RA-01	87-88	2.13	1.02	3,540	1,660	Central
RA-01-RC3-T2	BH-RA-01	92-93	2.13	1.07	2,920	1,300	Off-center
AC-01-RC1-T1	BH-AC-01	25-26	2.13	1.04	3,680	1,690	Central

ITS values range from 1,240 to 1,690 psi (mean 1,485 psi, std dev 160 psi). Peak loads range from 2,680 to 3,680 lbf. The UCS/ITS ratio of 15.1 is within the expected range for granodiorite (12 to 20). These values are adequate for the dam foundation and do not indicate any rock mass degradation.

3.4 Concrete Core Testing

3.4.1 Compressive Strength

Table 3-9: Concrete Core Compressive Strength (ASTM C42)

Specimen ID	Depth (ft)	D (in)	L (in)	Unit Wt. (pcf)	f _c (psi)	ASR Severity	Notes
AC-01-C1	2-4	3.75	7.50	148.5	4,850	Severe	Map cracking, gel in pores
AC-01-C2	6-8	3.75	7.48	149.2	5,920	Moderate	Reaction rims on agg.
AC-01-C3	10-12	3.75	7.52	150.1	6,480	Mild	Trace reaction rims
AC-01-C4	14-16	3.75	7.45	150.8	6,720	None	Sound concrete
AC-01-C5	18-20	3.75	7.50	151.2	6,890	None	Sound, near interior

The compressive strength results reveal a clear gradient from the exterior to the interior of the arch. The outermost 4 feet (severe ASR zone) retains a strength of 4,850 psi, which is approximately 21 percent above the original 28-day design strength of 4,000 psi but 26 percent below the expected aged strength of approximately 6,500 psi for sound concrete of this vintage. The interior concrete (below 12 feet depth) shows strengths of 6,480 to 6,890 psi, consistent with expected aged strength and confirming that ASR has not penetrated to the arch interior.

3.4.2 Split Tensile Strength

Table 3-10: Concrete Split Tensile Strength (ASTM C496)

Specimen ID	Depth (ft)	f _t (psi)	ASR Severity	Notes
AC-01-ST1	2-4	320	Severe	Failed along ASR crack network
AC-01-ST2	10-12	485	Mild	Clean tensile split
AC-01-ST3	18-20	510	None	Clean tensile split, sound

Tensile strength in the ASR-affected zone (320 psi) is 37 percent lower than the sound interior concrete (510 psi). This reduction in tensile capacity is the primary structural concern with ASR, as arch dams rely on tensile strength to resist thermal and seismic loads.

3.5 Petrographic Analysis (ASTM C856)

Petrographic examination was performed on four concrete core specimens at depths of 2, 6, 12, and 20 feet. The analysis confirmed the following:

- **ASR gel** was positively identified in specimens from 2 and 6 feet depth, occurring as deposits within aggregate particles (replacing reactive silica phases), along aggregate/paste interfaces, and filling cracks in both aggregate and paste. The gel is optically isotropic and exhibits the characteristic low birefringence of ASR product.
- **Reactive aggregate:** The coarse aggregate is predominantly rhyolite and dacite containing strained quartz, cryptocrystalline silica, and devitrified volcanic glass — all confirmed as alkali-reactive phases. The fine aggregate (natural sand) shows minimal reactivity.
- **Depth of ASR:** Active ASR (gel-filled cracks and reaction rims) was observed to a maximum depth of 6 feet from the downstream face. At 12 feet depth, only incipient reaction rims were observed on a few aggregate particles, with no gel formation or cracking. At 20 feet depth, no ASR indicators were present.

- **Carbonation:** Cement paste carbonation extends 0.8 inches from the exposed surface. This is a minor finding and does not affect structural performance.
- **Other degradation:** No evidence of freeze-thaw damage, sulfate attack, delayed ettringite formation, or reinforcement corrosion was observed.

3.5.1 Core Recovery and RQD

Table 3-11: Core Recovery and RQD Summary

Core Run	Boring	Interval (ft)	Material	Recovery (%)	RQD (%)	Notes
LA-01-RC1	BH-LA-01	72-85	Weathered grano.	65	18	Highly decomposed, friable
LA-01-RC2	BH-LA-01	85-100	Fresh granodiorite	94	78	Competent, 2 joint sets
LA-01-RC3	BH-LA-01	100-120	Fresh granodiorite	96	85	Excellent quality
RA-01-RC1	BH-RA-01	55-70	Weathered grano.	58	12	Very poor, grus zones
RA-01-RC2	BH-RA-01	70-82	Fresh granodiorite	91	72	Fair to good, tight joints
RA-01-RC3	BH-RA-01	82-95	Fresh granodiorite	93	82	Good quality
AC-01-C	BH-AC-01	0-22	Mass concrete	95	88	ASR cracking upper 4 ft
AC-01-RC1	BH-AC-01	22-28	Fresh granodiorite	97	92	Excellent, massive

4.0 SUBSURFACE CONDITIONS

4.1 Left Abutment

The left abutment boring confirms a 65-foot-thick embankment overlying 7 feet of original ground alluvial deposits, 13 feet of weathered granodiorite (grus), and fresh granodiorite bedrock extending to the boring termination at 120 feet. The embankment clay core (25 to 45 feet) is intact with index properties matching the original construction records. The core/foundation contact at 65 feet contains a thin (0.5-inch) clay gouge layer which appears to be the original stripped surface. No void, erosion channel, or piping indicator was observed at the contact.

The weathered granodiorite zone (72 to 85 feet) is the critical finding. This zone was identified in the 1964 design investigation and was treated with a grout curtain during construction (1967). However, the elevated piezometric pressure measured at 95 feet (18.2 psi = 42 ft head) indicates that the grout curtain may have deteriorated or been bypassed by seepage through the weathered zone. The weathered rock has UCS values of only 1,820 to 4,100 psi (compared to 18,400 to 25,600 psi for fresh rock) and RQD of only 18 percent, confirming it is a weak, permeable pathway.

4.2 Right Abutment

The right abutment boring confirms a 48-foot-thick embankment overlying 7 feet of dense glacial till (GC), 15 feet of weathered granodiorite, and fresh granodiorite to 95 feet. The embankment clay core (15 to 32 feet) is intact. The glacial till foundation (48 to 55 feet) is very dense with SPT N-values exceeding 70, consistent with the original construction records which described it as "very dense" and accepted it as competent foundation material.

The triaxial testing of the glacial till revealed measurable creep strain at high deviator stress levels. This finding, combined with the constant-rate displacement measured by survey monuments (0.2 inches per year over 25 years), supports the conclusion that the right abutment displacement is caused by long-term creep in the glacial till foundation under the sustained embankment load. The displacement is not indicative of progressive failure; rather, it represents a viscoplastic response of the till to sustained shear stress.

4.3 Arch Foundation

The arch concrete boring confirms 22 feet of mass concrete in good condition (aside from ASR in the outer 6 feet) overlying fresh granodiorite of excellent quality (RQD 92 percent, UCS 24,200 psi). The concrete/rock contact is intact with sound grout. The arch foundation is performing as designed and requires no remedial action.

5.0 ENGINEERING ANALYSIS

5.1 Seepage Assessment — Left Abutment

The seepage increase from 12 gpm (2012) to 28 gpm (2025) represents a 2.3-fold increase over 13 years. The investigation findings indicate that the seepage path is through the weathered granodiorite zone (72 to 85 feet depth) in the left abutment foundation, not through the embankment core. This conclusion is based on:

- The embankment clay core is intact with no evidence of piping, erosion, or void formation. Core permeability (estimated from grain size at 2.8×10^{-7} cm/s) is consistent with the design value and orders of magnitude lower than would be expected if piping were occurring.

- The elevated piezometric pressure at 95 feet (in fresh rock below the weathered zone) indicates that seepage is flowing through the rock foundation, not the embankment.
- The weathered granodiorite has RQD of only 18 percent and UCS of 1,820 to 2,650 psi, indicating a highly fractured and permeable zone. Original grouting records show relatively high grout takes in this zone (4 to 8 cf/LF), suggesting the grout curtain effectiveness may have been limited from the beginning or has deteriorated over time.
- The seepage water is clear with no turbidity, confirming no transport of fines from the embankment or foundation.

Recommendation: Supplemental grouting of the weathered granodiorite zone beneath the left abutment is recommended to reduce seepage and pore pressures. A grouting program consisting of a single row of grout holes at 10-foot spacing, drilled from the embankment crest to a minimum depth of 90 feet, with split-spacing as needed, should be designed. Estimated cost: \$180,000 to \$250,000.

5.2 Slope Stability — Right Abutment

An updated slope stability analysis was performed for the right abutment embankment using the laboratory-measured strength parameters and the current pore pressure distribution. The analysis used Spencer's method (ASTM D5321) with the following input parameters:

Table 5-1: Slope Stability Input Parameters

Material	Unit Wt. (pcf)	c' (psf)	phi' (deg)	Source
Gravel shell	140	0	38	Published values, verified by field density
Clay core	125	290	23	CU triaxial, BH-RA-01
Glacial till	145	450	34	CU triaxial, BH-RA-01
Weathered rock	150	800	35	Estimated from UCS correlation
Fresh granodiorite	165	N/A	N/A	Non-critical (below failure surface)

The calculated factors of safety are: **Static (normal pool): FS = 1.42; Seismic (MCE, pseudo-static k = 0.12): FS = 1.08; Rapid drawdown: FS = 1.28.** The static factor of safety exceeds the FERC minimum of 1.3 for normal loading conditions. The seismic factor of safety of 1.08 is below the FERC minimum of 1.1 for post-earthquake conditions, though it exceeds 1.0. A more refined seismic analysis (Newmark displacement method) is recommended to assess the acceptability of the seismic case.

The constant displacement rate (0.2 in/yr) and absence of accelerating pore pressures indicate that the embankment is not approaching a failure state. The displacement is attributable to creep in the glacial till foundation rather than a developing failure surface. Continued monitoring via the newly installed inclinometer and survey monuments is recommended. If the displacement rate increases or pore pressures begin to rise, a downstream buttress should be evaluated.

5.3 Concrete Structural Evaluation

The ASR has reduced concrete compressive strength by approximately 26 percent and tensile strength by approximately 37 percent in the outer 4 to 6 feet of the downstream face. However, the interior concrete (representing approximately 75 percent of the arch thickness) retains its full design strength. The structural significance of the ASR depends on the location and magnitude of tensile stresses in the arch under various loading conditions.

A preliminary screening analysis using the trial load method indicates that the maximum tensile stresses in the arch occur on the downstream face at the base of the dam under winter (empty reservoir, thermal contraction) loading. These stresses are estimated at 250 to 350 psi, which is within the tensile capacity of the ASR-affected concrete (320 psi from split tensile testing). However, the margin is very thin and will continue to decrease as ASR progresses.

Recommendation: A detailed finite element thermal-stress analysis of the arch incorporating the measured concrete property gradient (ASR-degraded exterior, sound interior) should be performed to assess structural adequacy under all loading combinations. In the interim, surface treatment of the downstream face with a penetrating lithium-based ASR inhibitor and/or a breathable surface coating should be evaluated to slow ASR progression. Estimated cost of surface treatment: \$120,000 to \$180,000.

6.0 CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

Based on the results of the PFR geotechnical investigation, MWDE concludes the following:

Left Abutment Seepage: The seepage increase is caused by flow through the weathered granodiorite foundation zone, not through the embankment core. The core is intact and performing as designed. The grout curtain effectiveness in the weathered zone has diminished, likely due to deterioration of grout in the highly fractured rock. Supplemental grouting is recommended.

Right Abutment Displacement: The displacement is caused by long-term creep in the glacial till foundation, not by a developing failure surface. The embankment and foundation are stable (FS = 1.42 static). The displacement rate is constant and not accelerating. Continued monitoring is recommended, with a contingency plan for a downstream buttress if conditions change.

Arch Concrete ASR: ASR has degraded the outer 4 to 6 feet of the downstream face, reducing compressive strength by 26 percent and tensile strength by 37 percent. The interior concrete (75 percent of arch thickness) is unaffected. The arch retains adequate structural capacity for current loading conditions but with reduced margin. Surface treatment and detailed structural analysis are recommended.

Foundation Rock: Fresh granodiorite foundation rock at all three boring locations shows strength and quality consistent with the 1964 design investigation data (mean UCS 22,467 psi vs. original 22,100 psi). The foundation has not degraded.

6.2 Recommended Actions

Table 6-1: Recommended Actions and Estimated Costs

Priority	Action	Estimated Cost	Timing
1	Left abutment supplemental grouting program	\$180K-\$250K	2026
2	Arch downstream face ASR surface treatment	\$120K-\$180K	2026-2027
3	Detailed FEM thermal-stress analysis of arch	\$60K-\$90K	2026
4	Right abutment inclinometer monitoring (quarterly)	\$8K/yr	Ongoing
5	Left abutment piezometer monitoring (monthly)	\$4K/yr	Ongoing
6	Newmark seismic displacement analysis (Rt. abt.)	\$25K-\$40K	2027
7	Contingency: Rt. abutment buttress design	\$350K-\$500K	If needed

7.0 LIMITATIONS

This report has been prepared for the exclusive use of the Elk Basin Irrigation District and its dam safety consultants for the specific project described herein. The findings are based on the subsurface conditions encountered at the three boring locations. Conditions may vary at other locations on the dam and appurtenant structures.

The borings reveal conditions only at the specific locations and depths explored. The embankment and foundation conditions between borings are inferred based on geologic interpretation, the original construction records, and professional judgment. If conditions different from those described in this report are encountered during future work, MWDE should be notified immediately.

The slope stability factors of safety presented in this report are based on deterministic analyses using the measured strength parameters. A probabilistic analysis or sensitivity study may yield different conclusions regarding the adequacy of the right abutment stability, particularly for the seismic loading case. The seismic analysis presented herein is a simplified pseudo-static screening; a more detailed analysis is recommended as noted in the report.

The concrete structural evaluation is a preliminary screening assessment. Definitive conclusions regarding the structural adequacy of the arch under all loading combinations, including thermal, seismic, and ASR-induced expansion, require a detailed finite element analysis which is beyond the scope of this investigation.